

June 12, 2026

Hiring Manager
Apex Systems

Subject: Application for Embedded Software/Firmware Engineer

Dear Hiring Manager,

I am applying for the Embedded Software/Firmware Engineer position at Apex Systems. My background in embedded firmware and secure IoT aligns well with your needs, particularly around ble, c, c++, can, embedded, firmware, linux, security.

Highlights from my experience:

- 15 years of embedded firmware experience across ARM Cortex-M, ESP32, PIC, and 8051
- Strong C/C++, RTOS/FreeRTOS, board bring-up, and device driver development
- Secure IoT delivery with MQTT/TLS, AES/ChaCha20-Poly1305, and post-quantum crypto implementation
- Hands-on systems integration with UART/SPI/I2C/RS-485/Modbus and cloud telemetry pipelines

I have led end-to-end firmware work from architecture and implementation through validation and field deployment, including reliability-focused telemetry systems and hardware-software integration on resource-constrained platforms. I am comfortable working across cross-functional teams to deliver production-ready embedded solutions.

I am currently based in Brookings, SD and open to relocation.

Thank you for your time and consideration. I would welcome the opportunity to discuss how my experience can contribute to your team.

Sincerely,
MANISH MAN SHRESTHA